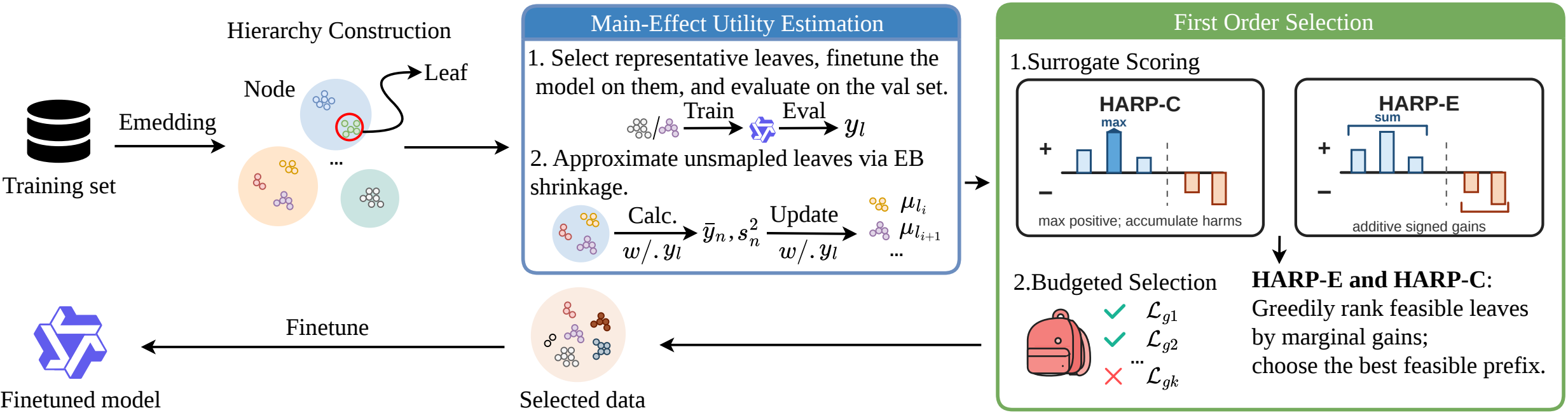

Hierarchy Construction
Node
Leaf
Emedding
Training set
Main-Effect Utility Estimation
1. Select representative leaves, finetune the model on them, and evaluate on the val set.
Train
Eval
2. Approximate unsmapled leaves via EB shrinkage.
Calc.
Update
First Order Selection
1.Surrogate Scoring
HARP-C
max
max positive; accumulate harms
HARP-E
sum
additive signed gains
2.Budgeted Selection
HARP-E and HARP-C:
Greedily rank feasible leaves by marginal gains;
choose the best feasible prefix.
Finetune
Finetuned model
Selected data